Modelling Resilience: Extension and Community-Based Adaptation in Kenya and Zimbabwe

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Abstract:
This study employs Structural Equation Modelling (SEM) to conduct a comparative analysis of agricultural extension systems and community-based climate adaptation strategies in Kenya and Zimbabwe, focusing on sweet potato farming. Both countries face escalating climate threats, prolonged droughts, erratic rainfall, and land degradation, but differ in institutional design and policy responses. Zimbabwe relies on a centralized extension service, while Kenya's decentralized framework empowers local leaders and community networks. Guided by the Sustainable Livelihoods Framework, Institutional Theory, and Resilience Theory, the research integrates expert insights to explore how extension policies, land tenure systems, and local governance structures influence the adaptive capacity of smallholder farmers. The findings underscore that successful climate adaptation hinges not only on technological innovation but also on institutional coordination, contextual knowledge, and grassroots inclusion. The study concludes with targeted policy recommendations to strengthen resilience and sustainability in agricultural extension systems across sub-Saharan Africa.

Keywords: *climate change, adaptation policies, Kenya and Zimbabwe, Structural Equation Modelling, sustainable livelihoods, farming systems.*

1. Introduction

Climate change presents a considerable systemic threat to global agriculture, disrupting weather patterns and intensifying extreme events such as droughts and floods. These changes jeopardize crop yields, threaten food security, and undermine the livelihoods of millions who depend on farming [1]. In Africa, where agriculture is predominantly rain-fed, the impacts are particularly severe, leading to decreased productivity and increased vulnerability among smallholder farmers [2]. Since 1961, the continent has experienced a 34% decline in agricultural productivity growth due to climate change—the highest globally [3].

Climate-resilient crops have emerged as vital tools in responding to these challenges. Adaptable to diverse agroecological zones, tolerant of extreme weather conditions, and often possessing short growing cycles, these crops—including sweet potatoes, millet, and cassava—support climate-smart agriculture in vulnerable farming systems [2,4,5]. Despite their resilience and nutritional value, they remain underutilized in mainstream policy, hindered by underinvestment and perceptions of being 'subsistence-only' [6,7].

Kenya and Zimbabwe provide contrasting but instructive contexts for exploring the integration of climate-resilient crops into national adaptation strategies. While both face escalating climate threats—prolonged droughts, erratic rainfall, and land degradation—they differ significantly in institutional design. Kenya's decentralized governance empowers county-level actors and community networks to tailor agricultural interventions, whereas Zimbabwe's centralized extension model provides national coherence but with limited local flexibility [8,9].

This study employs Structural Equation Modelling (SEM) to conduct a comparative analysis of agricultural extension systems and community-based climate adaptation strategies in Kenya and Zimbabwe, with a specific focus on the adoption and integration of climate-resilient crops like sweet potatoes. Through SEM, the study explores complex interrelationships between institutional quality, livelihood assets, and resilience outcomes.

Guided by the Sustainable Livelihoods Framework (SLF), Institutional Theory, and Resilience Theory, the research integrates expert insights to understand how extension services, land tenure systems, and local governance structures shape the adaptive capacity of smallholder farmers. Each of these theoretical lenses offers a unique perspective: SLF highlights the role of livelihood assets, Institutional Theory emphasizes formal and informal governance dynamics, and Resilience Theory focuses on the absorptive, adaptive, and transformative capacities of agricultural systems.

The study's objectives are fourfold:

1. To analyze the role of government and non-governmental extension services in supporting climate-resilient farming practices.
2. To evaluate the impact of land ownership patterns on the adoption of adaptive farming techniques and resilience strategies.
3. To assess the effectiveness of community-based approaches in building farmer capacity to address climate change challenges.
4. To examine policy frameworks influencing extension services and their alignment with climate adaptation goals.

By unpacking these dynamics, the study contributes to evidence-based policymaking and offers recommendations for enhancing the inclusivity, sustainability, and institutional coherence of agricultural adaptation frameworks in sub-Saharan Africa.

2. Study area

2.1. Goromonzi District, Zimbabwe: Agricultural Systems and Climate Adaptation

Geographical and Ecological Context Goromonzi District lies southeast of Harare in Zimbabwe's Mashonaland East Province. Spanning 25,407 km² and comprising 25 administrative wards—including communal, commercial, and small-scale farming zones—the district benefits from fertile soils, moderate temperatures (15–20°C), and annual rainfall averaging 800–1000 mm, mostly in the summer [10,11]. These features support diverse agricultural activities, though increasing rainfall variability presents significant challenges.

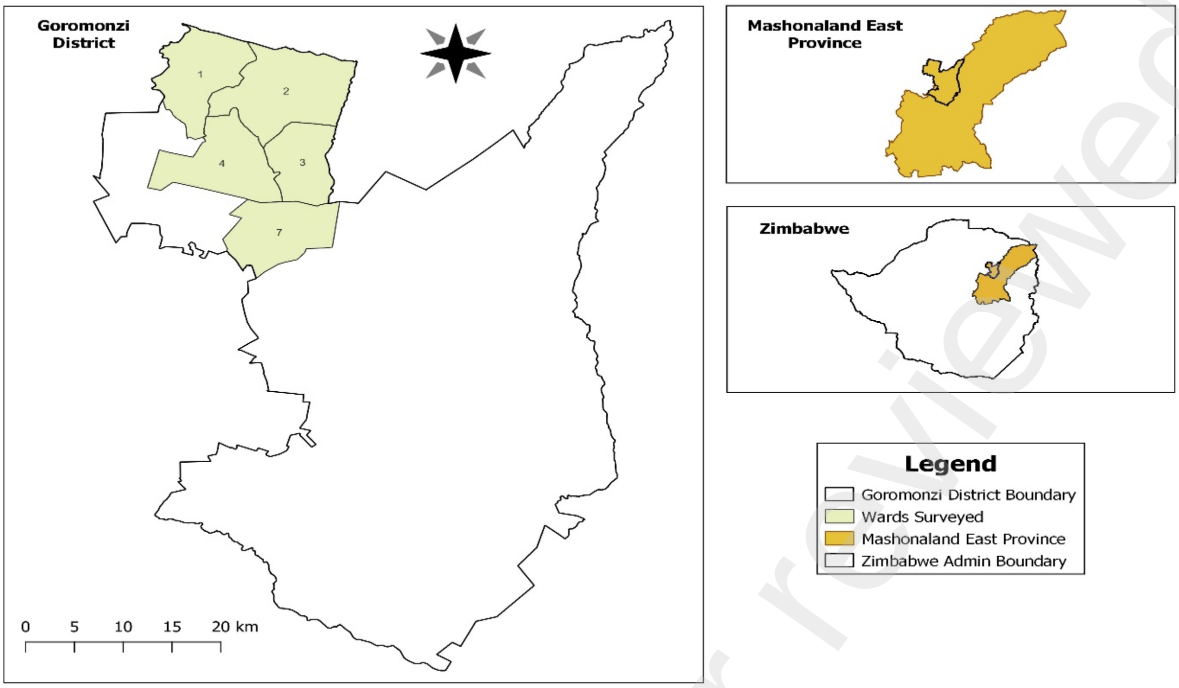

Fig 1. Goromonzi district and selected wards [12]

2.2 West Pokot County: Overview of Agricultural Systems and Climate Adaptation

Geographical and Ecological Context West Pokot County, in northwestern Kenya, straddles semi-arid and sub-humid zones of the Rift Valley. Despite ecological stressors like frequent droughts, erratic rainfall, and land degradation [13], the county's diverse agro-ecological zones—from highlands to lowland plains—enable varied and resilient farming practices. These features make West Pokot a suitable case for studying community-based adaptation strategies [14].

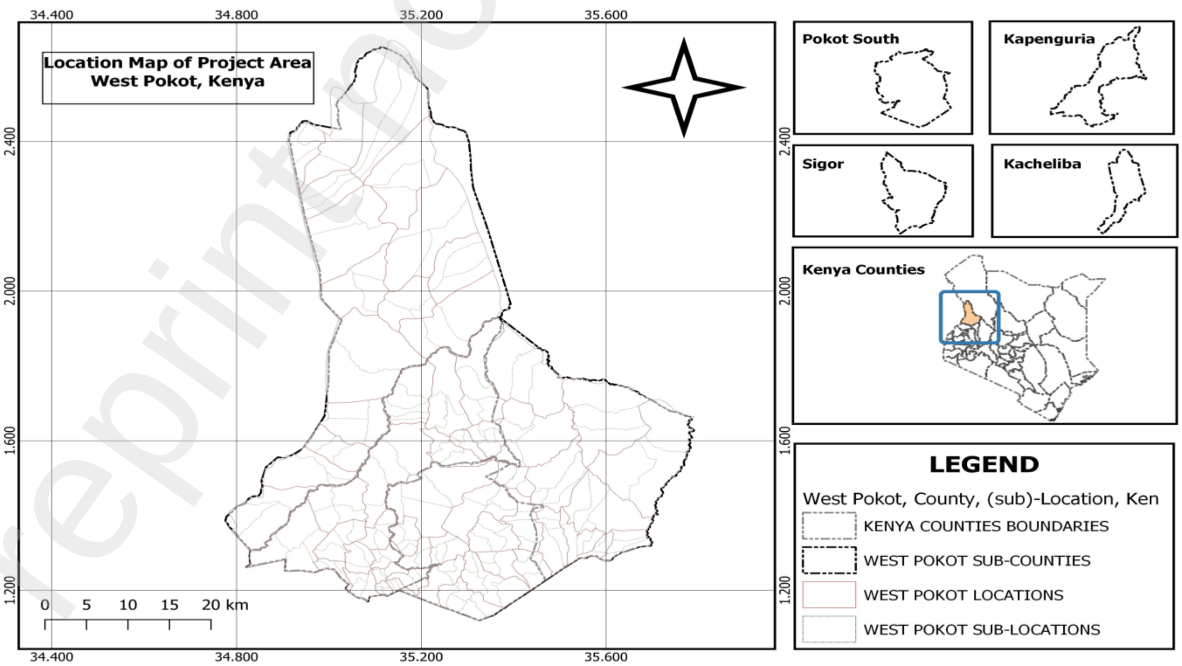

Fig. 2. West Pokot County, sub-counties, wards, locations, and sub-locations [15]

3. Methodology

3.1. Comprehensive Literature Review

The initial phase of this research undertakes a comprehensive literature review to establish the conceptual, empirical, and policy foundations for understanding the intersections between migration, climate resilience, and governance in Zimbabwe and Kenya. This review draws from a broad spectrum of sources, including peer-reviewed academic journals, policy documents, and reports from international organizations such as the FAO, IOM, UNDP, and the World Bank [2,16,17].

The literature review is organized around three interlinked theoretical pillars: the Sustainable Livelihoods Framework (SLF), Institutional Theory, and Resilience Theory. Through this structure, the review identifies how livelihoods are shaped and constrained by climate change, migration flows, and institutional responses in both countries.

Under the SLF, the literature focuses on the five capitals—natural, social, human, physical, and financial—and their role in shaping the vulnerability and adaptive capacity of smallholder farmers and rural migrants. Studies highlight how land degradation, climate variability, and limited infrastructure exacerbate livelihood fragility, while remittances, informal savings networks, and community support systems help buffer shocks [18,19]. Specific attention is paid to literature examining the agricultural sectors in West Pokot (Kenya) and Goromonzi (Zimbabwe), where rain-fed farming systems dominate and are increasingly pressured by climatic and socio-economic shifts [9,20].

In examining Institutional Theory, the review surveys both formal policy frameworks—including land tenure systems, extension services, and climate adaptation plans—and informal institutions, such as customary land governance and traditional leadership structures. Key themes include the hybrid governance models present in both countries, the interplay between state and customary authority, and the institutional bottlenecks that hinder access to resources or the implementation of inclusive adaptation strategies [8,21]. Literature on legal pluralism and decentralization also informs the analysis of how different governance regimes shape adaptation capacity and policy responsiveness in marginalized communities.

With respect to Resilience Theory, the review investigates how social and ecological systems respond to shocks and stressors. It categorizes resilience into three dimensions: absorptive, adaptive, and transformative capacity [22]. The review synthesizes literature on climate-smart agriculture, disaster risk reduction, and institutional innovation, especially in regions characterized by recurring droughts, conflict over resources, and limited public investment. Case studies from both Kenya and Zimbabwe offer insights into how localized interventions—such as agroecological practices, community-based disaster response, and participatory planning—contribute to building long-term resilience [23,24].

Moreover, the literature review incorporates comparative studies that draw parallels between Kenya, Zimbabwe, and other climate-affected countries in Africa. These comparisons help contextualize national experiences within broader regional trends and inform cross-case learning on climate adaptation and migration governance. In doing so, the review also evaluates the influence of regional frameworks such as the African Union Migration Policy Framework and international agreements like the Global Compact for Safe, Orderly and Regular Migration [25,26].

Finally, the literature review identifies gaps in the existing knowledge base—particularly the need for integrated, multi-scalar analysis that links household-level resilience with national policy and institutional dynamics. These gaps inform the study's methodological choices and highlight the importance of merging quantitative data with expert and community-based insights.

3.2. Theoretical Framework

3.2.1. Sustainable Livelihoods Framework (SLF)

This study adopts the Sustainable Livelihoods Framework (SLF) to examine how smallholder farmers in Kenya and Zimbabwe adapt to climate change within rain-fed agricultural systems. The SLF is a people-centered analytical tool that emphasizes how individuals, households, and communities mobilize and combine various types of assets—natural, social, human, physical, and financial capital—to sustain their livelihoods in the face of environmental, economic, and institutional stressors [27,19]. The framework facilitates a holistic understanding of how community agency and institutional support interact to shape the resilience and adaptive capacity of smallholder agricultural systems.

In the context of climate change adaptation, SLF enables an exploration of both the constraints and opportunities that influence farmers' responses to increasing climatic variability. It is particularly suited to Kenya and Zimbabwe, where livelihood strategies are deeply embedded in local ecologies, policy contexts, and socio-economic structures.

3.2.1.1. Natural Capital

Natural capital includes access to productive land, water, soil fertility, and ecosystem services—all essential resources for agriculture. In both Kenya and Zimbabwe, smallholder farmers depend heavily on rain-fed agriculture, which is increasingly undermined by erratic rainfall patterns, prolonged droughts, and land degradation [28].

In West Pokot County, located in Kenya's arid and semi-arid lands (ASALs), natural capital forms the cornerstone of smallholder farming and pastoral livelihoods. However, the region faces acute challenges related to declining water availability, land degradation, and soil fertility loss, which are exacerbated by climate change, population pressure, and unsustainable land use practices. Erratic rainfall patterns, increasing frequency of droughts, and seasonal river variability have further undermined the reliability of natural resources upon which local farmers depend [29].

Despite these vulnerabilities, several community-led and government-supported interventions have emerged to rebuild natural capital and enhance climate resilience. For instance, agroecological zoning and land-use planning introduced through devolved county governance structures have helped identify appropriate areas for crop farming, grazing, and conservation, reducing pressure on fragile ecosystems. Additionally, community-based water harvesting techniques—such as sand dams, shallow wells, and roof catchment systems—have been promoted by NGOs and county programs to stabilize water access for both agricultural and domestic use. These efforts not only replenish water reserves but also reduce conflicts over water and improve dry season crop productivity.

Furthermore, integrated approaches such as soil and water conservation (SWC) through terracing, agroforestry, and use of organic inputs have helped restore degraded land and build long-term fertility. These practices, often adopted through farmer field schools and local extension initiatives, contribute to sustainable land management while preserving the ecological base for future agricultural use.

Yet, access to and control over natural capital remains uneven. Land tenure insecurity, especially in areas with customary land systems, can discourage investment in long-term conservation efforts, limiting the impact of these initiatives. Thus, while West Pokot provides a promising example of community-driven restoration of natural capital under climate stress, its success depends heavily on the integration of local knowledge, inclusive policies, and supportive institutions.

In Goromonzi District, located in Zimbabwe's Mashonaland East Province, smallholder farmers rely heavily on natural capital—especially land and rainfall-dependent water resources—for crop and livestock production. Following Zimbabwe's Fast Track Land Reform Programme (FTLRP), land redistribution in Goromonzi significantly altered land access dynamics, transferring large-scale commercial farmland into the hands of smallholder and communal farmers. While this restructured land access and provided new livelihood opportunities for previously marginalized populations, the resultant communal and leasehold tenure systems have presented significant constraints to the long-term development and sustainability of natural capital [30].

In areas of communal tenure, many farmers lack secure, documented rights to land, which deters them from making long-term investments in soil fertility management, irrigation, and erosion control. This insecurity is compounded by environmental stresses: Goromonzi is increasingly experiencing rainfall variability, including both seasonal droughts and flash floods, which degrade soil structure, increase erosion, and threaten food security. Without clear land rights or long-term guarantees, farmers are less inclined to engage in practices such as terracing, agroforestry, or water harvesting—all of which are crucial for regenerating natural capital under climate stress.

Nevertheless, some farmers in Goromonzi have adopted adaptive land-use practices—particularly those supported by extension agents, NGOs, and donor-funded projects. Conservation agriculture, which involves minimum tillage, mulching, and crop rotation, has shown promise in improving soil structure and moisture retention, especially when applied on formerly overworked soils. In some cases, these efforts are linked with climate-smart agriculture initiatives such as Pfumvudza/Intwasa, which promote intensive land preparation on small plots using organic matter and moisture-conserving techniques.

Yet, the full potential of these interventions is limited without secure land tenure and consistent institutional support. Customary tenure arrangements in communal areas—where land is allocated by traditional leaders—often operate without written records or legal enforceability. This undermines farmers' ability to use land as collateral or to access credit for purchasing soil amendments, fencing, or irrigation equipment, thereby constraining the development of natural capital.

In sum, while Goromonzi offers examples of promising community responses to land degradation and climate stress, the effectiveness of natural capital restoration depends largely on resolving land tenure insecurity, scaling up farmer training, and integrating localized environmental knowledge into formal policy and extension programs.

3.2.1.2. Social Capital

Social capital comprises the networks, relationships, norms of trust, and collective action within a community is a key component of the Sustainable Livelihoods Framework (SLF) and plays a vital role in enabling smallholder farmers to adapt to climate change. In both Goromonzi District in Zimbabwe and West Pokot County in Kenya, communities have developed grassroots institutions such as community-based organizations (CBOs), savings and credit groups, and agricultural cooperatives, which act as crucial vehicles for organizing and implementing adaptation strategies.

In Goromonzi, social capital is deeply embedded in communal relationships shaped by shared land histories, traditional leadership structures, and localized responses to economic instability and climate risks. In the absence of robust state support, CBOs and informal savings groups—often known as Internal Savings and Lending (ISAL) groups—serve as social safety nets for women and marginalized households, allowing members to pool resources for purchasing drought-tolerant seeds, fertilizer, or livestock feed. These groups not only facilitate resource sharing and risk pooling, but also enable collective decision-making around land use, planting times, and conservation strategies in response to seasonal variability.

Additionally, church groups and farmers' clubs—many supported by NGOs—act as knowledge-sharing platforms for climate-smart practices such as conservation agriculture, mulching, and water harvesting. These social institutions foster trust and collaboration, which are essential for spreading innovation and mobilizing labor for community-based projects like gulley reclamation or tree planting. However, social capital in Goromonzi can be limited by politicization of community leadership and exclusionary practices, particularly in areas where traditional leaders are closely aligned with political elites, affecting who benefits from donor and government support [20].

In West Pokot, social capital is a lifeline in a region prone to drought, food insecurity, and livestock losses. Ethnic solidarity, clan affiliations, and traditional institutions such as elders' councils play key roles in conflict resolution, natural resource management, and coordinated responses to environmental stress. At the same time, modern CBOs, women's savings groups (chamas), youth self-help groups, and farmer cooperatives have emerged as effective mechanisms for organizing climate adaptation efforts at the local level.

For instance, chamas enable members—often women—to save collectively, access microloans, and invest in small-scale farming or water storage systems. Cooperatives in the region have also promoted livestock vaccination campaigns, early warning dissemination, and fodder banking for drought preparedness. These institutions often partner with county government climate platforms and NGOs to scale up climate action plans and facilitate community-based resilience building, especially in semi-arid zones.

Importantly, devolution in Kenya has strengthened the role of community groups by allowing greater participation in county-level planning and budgeting, especially through public participation forums [8]. This institutional link enhances the legitimacy and impact of grassroots initiatives and ensures local needs are reflected in formal adaptation policies.

3.2.1.3. Human Capital

Human capital—comprising knowledge, skills, education, and health—is a vital component of farmers' ability to respond to climate risks, adopt new technologies, and engage meaningfully in climate-resilient agriculture. In the context of smallholder farming systems, strong human capital underpins the effectiveness of adaptation strategies, influencing how well communities can access, interpret, and apply agro-climatic information.

In West Pokot, there has been notable progress in expanding access to climate-smart knowledge and agricultural training, especially through the integration of mobile-based platforms such as M-Farm and iCow. These platforms deliver real-time agro-advisories, market prices, weather forecasts, and agronomic tips directly to farmers via mobile phones, thus reducing dependency on physical extension officers and enhancing knowledge dissemination across geographically dispersed communities [31]. Such innovations have proven especially useful in overcoming barriers related to infrastructure, literacy, and extension coverage, and are being increasingly adopted by youth and women involved in farming.

In addition, farmer field schools (FFSs) and community demonstration plots—often facilitated by NGOs and government extension services—have enhanced practical learning around topics like soil conservation, crop diversification, and integrated pest management. Through these platforms, farmers not only acquire technical skills but also develop a culture of peer-to-peer learning and experimentation, which is critical for local innovation and adaptation. However, disparities remain, with pastoralist zones in the lowlands often having more limited access to formal education and agricultural training, necessitating further investment in localized and culturally appropriate education models.

In contrast, Goromonzi District faces more pronounced challenges in the development of human capital, particularly within communal areas and resettled communities. Many rural farmers have limited access to extension services, climate education, and technical training, which constrains their ability to adopt sustainable practices or respond effectively to changing environmental conditions [9]. The delivery of agricultural training is often constrained by resource shortages, staff capacity gaps, and the underfunding of public institutions. This situation has been worsened by Zimbabwe's broader economic challenges, which have led to a decline in public service delivery and investment in rural education infrastructure.

Additionally, access to climate information and digital platforms is significantly lower compared to Kenya. Few farmers in Goromonzi use mobile phones for accessing weather forecasts or Agro-advisories due to network limitations, high data costs, and digital literacy gaps. In marginalized zones, especially among women and elderly farmers, this digital divide further limits exposure to climate-smart agriculture training and innovation.

Nonetheless, some progress has been made through NGO-led initiatives and farmer cooperatives, which offer targeted training on conservation agriculture, drought-tolerant crop varieties, and post-harvest handling. However, without systemic investment in education, training infrastructure, and digital connectivity, human capital in Goromonzi remains a limiting factor in building widespread resilience.

3.2.1.4. Physical Capital

Physical capital refers to the basic infrastructure and productive assets that support livelihood activities, including transport networks, irrigation systems, storage facilities, communication infrastructure, and market access points. In the context of climate-resilient agriculture, physical capital plays a critical role in enabling adaptation, reducing transaction costs, and linking smallholder farmers to extension services, input markets, and buyers.

In Goromonzi, access to physical capital remains uneven and often inadequate, particularly in communal areas, where infrastructure development lags behind commercial farming zones. Decades of economic instability, underinvestment, and currency volatility have led to the deterioration of roads, irrigation schemes, storage depots, and electricity infrastructure, severely limiting agricultural productivity and market connectivity [32].

The inadequacy of physical infrastructure presents a major barrier to agricultural productivity and climate resilience, particularly for communal and smallholder farmers. Many of the rural roads remain unpaved, and during the rainy season, they become largely impassable, severely restricting farmers' ability to access markets and transport produce in a timely manner [32]. Furthermore, irrigation systems, which were largely developed during the commercial farming era, have deteriorated due to lack of maintenance and investment, rendering them largely inaccessible to smallholder and communal farmers who are most vulnerable to rainfall variability [30]. In addition to these challenges, post-harvest storage facilities are limited, leading to significant spoilage and food loss, especially during periods of surplus when infrastructure for drying, cooling, or storage is most critical [24]. These physical capital deficits not only constrain agricultural output but also hinder farmers' ability to adopt climate-resilient practices and improve their economic standing.

Despite these challenges, various NGO-led and donor-supported initiatives are working to improve physical capital in targeted communities through the rehabilitation of small dams, promotion of low-cost irrigation technologies, and establishment of community seed banks and solar-powered boreholes. However, without sustained public investment and broader policy reforms, disparities between commercial and communal farming zones will continue to limit equitable resilience outcomes in Goromonzi.

In contrast, West Pokot has benefited from Kenya's devolved governance system, which has facilitated more targeted investments in rural infrastructure through county-level planning and budgeting [29]. While infrastructure gaps persist, especially in arid and remote areas, notable progress has been made in improving market access, rural electrification, water infrastructure, and transport connectivity.

Investments in physical capital have played a significant role in improving agricultural productivity and climate resilience. The expansion of feeder roads in highland areas has greatly enhanced farmers' ability to access regional markets, thereby reducing post-harvest losses and improving the distribution of food across local and external supply chains [29]. In pastoral and agro-pastoral zones, water infrastructure projects—including earth dams, water pans, and boreholes—have significantly improved access to water for both irrigation and livestock, mitigating the impacts of prolonged dry spells [14]. Moreover, through collaborations with NGOs and national programs, the county has actively supported the adoption of rainwater harvesting systems, shallow wells, and solar-powered drip irrigation technologies, particularly in drought-prone communities. These integrated initiatives have strengthened local adaptive capacity by supporting year-round farming and reducing dependence on erratic rainfall [23]. Moreover, county governments and NGOs have collaborated to establish livestock markets, farmer aggregation centers, and storage facilities for grains and animal feed, helping improve value chain efficiency. The integration of digital infrastructure, such as mobile platforms for market information and e-extension services, further complements physical capital development, although signal and network coverage remain a challenge in certain zones.

3.2.1.5. Financial Capital

Financial capital refers to the financial resources that people use to achieve livelihood objectives. This includes savings, credit, remittances, access to financial institutions, and informal or digital financial systems. In the context of smallholder agriculture and climate resilience, access to reliable and affordable financial capital is essential for investing in inputs, adopting adaptive technologies, and recovering from climate-induced losses.

In Goromonzi, smallholder farmers—particularly those in communal areas—face significant constraints in accessing financial capital. These constraints are shaped by macroeconomic instability, hyperinflation, and a fragile financial sector, all of which undermine farmers' ability to plan, invest, or save effectively [20].

Access to formal financial capital remains a major barrier for smallholder and communal farmers, significantly limiting their capacity to invest in climate-resilient agricultural practices. Formal credit facilities are largely inaccessible, primarily due to the lack of collateral**,** high interest rates, and stringent borrowing requirements that many rural farmers cannot meet. This challenge is especially acute in communal areas, where most farmers lack titled deed**s**, a requirement for securing loans from conventional banks [20]. Additionally, hyperinflation and ongoing currency volatility in Zimbabwe discourage long-term savings and undermine trust in financial institutions. As a result, many banks have scaled back rural operations, citing high operational risks and liquidity constraints [24]. The continued dominance of a cash-based economy, coupled with unreliable mobile money platforms, further restricts financial inclusion, particularly for farmers in remote or under-served communities. These conditions create a financial environment that hinders not only day-to-day farming operations but also the long-term investments needed for adaptation and productivity enhancement.

In response, many farmers rely on informal financial systems such as Internal Savings and Lending (ISAL) groups, rotating savings groups, and church-based lending schemes. These locally organized platforms provide an alternative mechanism for pooling capital, accessing emergency funds, and supporting small investments in tools, seeds, or fertilizers. While useful, these systems are often constrained by limited fund sizes and vulnerability to broader economic shocks.

Donor-supported projects and NGOs have attempted to fill these gaps by offering grant-based support or subsidized finance through input voucher programs and revolving funds, especially targeting women and youth. However, the scale and sustainability of such interventions remain limited in the absence of a stable macro-financial environment.

In contrast, West Pokot has seen a notable transformation in financial inclusion due to the expansion of digital financial services. Platforms like M-Pesa (mobile money transfer) and M-Shwari (digital savings and microcredit) have significantly improved access to credit and savings, even in remote and underserved rural communities [33].

The rapid expansion of mobile financial services has significantly enhanced financial inclusion and resilience among smallholder farmers. Through platforms such as M-Pesa and M-Shwari, farmers are able to save small amounts of money securely, access microloans without formal collateral, and conduct cashless transactions to purchase agricultural inputs and services [33]. These mobile-based solutions reduce reliance on traditional banking infrastructure, which is often limited or inaccessible in remote rural areas, thereby enhancing financial resilience, especially during lean agricultural seasons or in the aftermath of climate-induced shocks such as droughts or livestock losses. Furthermore, the uptake of mobile insurance products, such as Kilimo Salama, is steadily increasing. These weather-indexed insurance schemes offer farmers compensation for crop failure triggered by adverse climatic conditions, thereby enabling them to recover more quickly and maintain productivity [23]. As a result, mobile finance has become a cornerstone of adaptive capacity in rural Kenya, supporting both livelihood security and long-term resilience.

Additionally, local SACCOs (Savings and Credit Cooperative Organizations), village savings and loan associations (VSLAs), and women's chamas (rotating savings groups) continue to provide crucial financial support, particularly for marginalized populations with limited digital literacy or mobile phone access.

The county government and NGOs have also promoted financial literacy programs and facilitated partnerships between farmers and financial service providers. These efforts aim to ensure that access to finance goes hand-in-hand with responsible borrowing and productive investment in climate-smart practices such as drought-tolerant seeds, water harvesting systems, and improved livestock breeds.

3.2.2. Institutional Theory

Institutional Theory provides a valuable lens to examine how the structures, rules, and norms that govern agricultural systems shape the behavior of smallholder farmers and their ability to adapt to climate change. Institutions—both formal (e.g., government policies, legal frameworks, and extension systems) and informal (e.g., cultural practices, social norms, and traditional authority)—play a central role in mediating access to resources, determining land tenure arrangements, and shaping responses to climate-related risks.

In the climate-stressed, rain-fed agricultural systems of Kenya and Zimbabwe, Institutional Theory helps us understand why some smallholder communities are more adaptive and resilient than others, despite facing similar environmental conditions. It allows us to evaluate how institutional arrangements enable or constrain community-based adaptation, the functioning of extension services, and land governance.

3.2.2.1. Formal Institutions and Access to Resources

In Goromonzi, formal institutions such as the Ministry of Lands, Agriculture, Water, Fisheries and Rural Development, along with Agricultural and Rural Development Advisory Services (ARDAS), are tasked with providing extension services, implementing agricultural policy, and supporting land governance. However, the effectiveness of these formal structures is often constrained by limited financial resources, staff shortages, and centralized governance that fails to account for localized needs [9]. While government-led climate initiatives such as the Pfumvudza program aim to promote conservation agriculture, uptake in communal areas is inconsistent due to lack of consistent support and secure land tenure.

Formal land tenure systems in Goromonzi include state leasehold, freehold, and customary tenure, but most smallholder and communal farmers lack legal title, which severely restricts their access to credit and formal resource support [20]. As a result, even where climate-smart practices are promoted, the incentive and capacity to adopt them are limited by structural exclusion and tenure insecurity.

In West Pokot, formal institutions have been revitalized through devolution**,** which shifted agricultural, water, and land governance to county-level authorities. This shift has enabled more targeted, participatory, and flexible responses to climate-related challenges. The County Department of Agriculture, working with national agencies like KALRO, has developed climate action plans and extension programs tailored to local agro-ecological realities [29].

Despite the progress, challenges persist. Many formal services still face capacity limitations**,** particularly in remote or conflict-prone areas. However, unlike in Zimbabwe, Kenya's policies—such as the Climate Change Act (2016) and Community Land Act (2016)—provide frameworks for integrating local institutions and recognizing communal land rights, which enhances the legitimacy and effectiveness of formal adaptation efforts [8].

3.2.1.2. Informal Institutions and Land Tenure Governance

In Goromonzi District, informal institutions such as traditional leadership structures—including chiefs, headmen, and village elders—play a central role in mediating access to communal land, resolving intra-community disputes, and overseeing the allocation of agricultural plots. These leaders are often the primary point of contact between rural communities and external actors, including government agencies, NGOs, and agricultural extension services. In the absence of well-functioning formal land administration, traditional authorities shape how resources are distributed and who benefits from development interventions [20].

These roles are reinforced by long-standing cultural practices and social norms, such as respect for seniority, ancestral land rights, and patrilineal inheritance systems. In many communal areas, land is viewed not as an individual commodity but as a collective asset linked to family lineage and cultural identity. This cultural framing influences how farmers perceive tenure security and shapes their willingness to invest in long-term land improvements such as irrigation, agroforestry, or soil conservation [34].

However, the authority of traditional leaders is not without challenges. Their influence is occasionally undermined by political patronage, elite manipulation, and exclusionary practices that favor certain families or political affiliations, especially in areas impacted by land reform or resource conflicts. Women and youth, in particular, may face barriers to land access due to entrenched patriarchal norms and a lack of recognition of their customary land rights [21]. Moreover, without formal integration into national legal frameworks, these customary arrangements can reinforce ambiguity and inhibit farmers from accessing credit or participating in government-supported adaptation programs that require proof of land ownership.

In West Pokot, customary institutions such as councils of elders, age-set systems, and clan-based land management practices are deeply embedded in community governance. These informal institutions are widely respected and play essential roles in natural resource management, particularly in managing communal grazing lands, resolving land disputes, and coordinating rotational grazing systems that prevent land degradation [23]. Their legitimacy is grounded in oral traditions, cultural rituals, and collective memory, which guide decision-making and promote community compliance with shared rules.

These institutions operate alongside formal county structures, and their roles are increasingly acknowledged in Kenya's legal and policy frameworks, particularly through the Community Land Act [35]. This legislation enables communities to register communal land, thereby bridging the gap between customary and statutory land governance and increasing the potential for community-driven adaptation strategies [8].

Cultural practices such as collective cattle herding, seasonal mobility, and ritual consultations with elders before major land use decisions illustrate how social norms and belief systems influence climate adaptation. For instance, during times of drought, elders may declare temporary migration routes or restrict grazing zones—decisions rooted not just in ecological reasoning but also in customary law and cultural legitimacy. These informal mechanisms ensure local ownership of adaptation strategies, reduce conflict, and foster social cohesion, which is essential for managing shared resources in climate-vulnerable environments.

Importantly, gender norms also shape participation in these institutions. While male elders traditionally dominate decision-making processes, there is growing recognition of the role of women's groups, especially chamas (informal savings and investment groups), in managing water resources, promoting reforestation, and leading food security initiatives [24]. This evolving dynamic highlights the transformative potential of informal institutions when they adapt to changing social roles and integrate marginalized voices.

3.2.3. Resilience Theory

Resilience Theory provides a valuable framework for analyzing how systems—social, ecological, and agricultural—respond to disturbances such as climate change. Rather than focusing solely on vulnerability or risk, resilience theory emphasizes the capacity of systems to absorb shocks, adapt to change, and transform in response to long-term stresses such as climate change. In the context of smallholder agriculture, this includes the ability of farmers and institutions to withstand droughts, floods, crop failure, and other climate-induced stresses, while continuing to sustain food production, income generation, and ecological integrity.

3.2.3.1. Absorptive Capacity: Withstanding Climate Shocks

Absorptive capacity refers to the immediate ability of systems—social, ecological, or economic—to withstand the effects of sudden climate shocks such as droughts, floods, or heatwaves, without undergoing significant disruption to their core functions or structures. It encompasses access to safety nets, basic infrastructure, early warning systems, and risk-sharing mechanisms that help individuals and communities absorb stress and minimize damage in the short term.

In Goromonzi, absorptive capacity is relatively weak, particularly in communal farming areas that lack the infrastructure and institutional support needed to respond effectively to sudden climate disruptions. Recurring droughts, increasingly erratic rainfall, and the progressive degradation of natural resources—combined with aging or underdeveloped infrastructure—have significantly eroded local systems' ability to buffer against shocks (Mugandani et al., 2021). Many farmers in Goromonzi lack crop insurance, irrigation systems, or post-harvest storage facilities, leaving them exposed to crop failures and income losses during adverse climatic events.

Additionally, few households have sufficient food reserves, savings, or access to emergency credit, which means that even moderate shocks can trigger cycles of food insecurity and debt. While community-based coping strategies such as food sharing, informal lending, and communal labor exist, these mechanisms are often stretched to their limits during prolonged crises or when multiple households are simultaneously affected (Mujeyi et al., 2022). The absence of localized early warning systems, disaster preparedness training, or accessible government relief also limits timely response, increasing dependence on external aid or short-term coping behaviors that may compromise long-term sustainability.

Moreover, the limited reach of extension services and under-resourced local institutions constrain farmers' access to drought mitigation knowledge and technologies. As such, Goromonzi's current absorptive capacity reflects a reactive posture, with many households unable to shield themselves effectively from immediate climate-induced disruptions.

In contrast, West Pokot exhibits relatively stronger absorptive capacity, particularly in areas where community cohesion, traditional knowledge, and decentralized governance intersect to support local resilience. Communities have long adapted to semi-arid conditions through livestock mobility strategies, enabling households to relocate animals to areas with better water or forage during dry spells—an indigenous practice that remains central to shock absorption [23]. These systems are often managed through clan networks and elders' councils, which enforce grazing rules and coordinate access to shared resources.

The development of water harvesting infrastructure, including earth dams, rock catchments, water pans, and boreholes, further supports immediate resilience during periods of rainfall failure. Many of these facilities have been constructed or rehabilitated through county government investments and NGO partnerships, demonstrating the value of collaborative governance in enhancing local absorptive capacity [36].

In addition, the widespread use of digital financial platforms like M-Pesa has improved household liquidity, allowing farmers to receive remittances, access emergency credit, and make cashless transactions even during climate crises. Some households also participate in village savings and loan associations (VSLAs) or insurance schemes (e.g., Kilimo Salama), which provide a safety net in times of need [33]. Importantly, the institutional integration of community-based disaster risk management into county planning allows for timelier and context-specific response strategies, such as early livestock vaccination campaigns, food relief distribution, and emergency water trucking during droughts.

However, absorptive capacity in West Pokot is not uniform. In more remote or marginalized zones, infrastructural gaps, weak market access, and recurrent conflict over resources can still undermine short-term resilience. Nonetheless, the synergy between indigenous practices, digital innovation, and localized governance creates a more dynamic system capable of withstanding climatic stress without widespread disruption.

3.2.3.2. Adaptive Capacity: Adjusting to Climatic Variability

Adaptive capacity refers to a system's ability to adjust to climate-related changes through learning, experimentation, innovation, and flexible management of resources. It reflects the potential of individuals, households, and institutions to make informed choices, reorganize livelihoods, and apply new knowledge in response to both gradual and sudden environmental shifts. In agricultural settings, adaptive capacity is particularly important for coping with long-term stressors such as prolonged droughts, erratic rainfall, soil degradation, and shifting growing seasons.

In Goromonzi, adaptive capacity is emerging, but remains highly uneven across communities. The introduction and partial uptake of conservation agriculture (CA)—which emphasizes minimum tillage, mulching, crop rotation, and soil cover—has helped some smallholders improve soil health and moisture retention, thereby mitigating the effects of rainfall variability (Mango et al., 2017). In addition, farmers in select areas have begun adopting drought-tolerant crop varieties, such as orange-fleshed sweet potatoes, cowpeas, and small grains, which are better suited to changing climatic conditions [20].

Small-scale irrigation schemes, especially those established through donor or NGO support, have also helped households diversify their cropping cycles and extend production into drier months. However, the success and sustainability of these adaptations are constrained by insecure land tenure, particularly in communal areas where customary land rights are not formally recognized. Without security of tenure, farmers are reluctant to invest in long-term adaptive infrastructure such as irrigation, agroforestry, or soil reclamation systems.

Moreover, access to agricultural extension services remains limited. Public services are underfunded and overstretched, while NGO-led interventions often depend on short-term project cycles and donor funding, which can limit the continuity and scalability of adaptation support. Additionally, a lack of access to finance, including credit, input subsidies, and insurance, hampers farmers' ability to take risks or invest in adaptive innovations. In areas where external actors provide training and inputs, adaptive capacity is noticeably higher, but these benefits are not evenly distributed across wards, reinforcing regional disparities.

In contrast, West Pokot has made significant strides in institutionalizing adaptive capacity, largely due to the devolution of governance under Kenya's 2010 Constitution. County-level planning has enabled more context-responsive and participatory adaptation strategies, aligning local knowledge with formal extension and climate policy frameworks. As a result, adaptive strategies are not only introduced but are increasingly integrated into daily agricultural practices.

Farmers in West Pokot employ crop diversification to spread climate risk across multiple food and income sources. This includes combining cereals with drought-resistant crops such as sorghum, millet, and indigenous legumes. In agro-pastoral zones, there is growing adoption of agroforestry systems, which combine trees, crops, and livestock to improve soil fertility, reduce erosion, and provide year-round income from products like fruit, fodder, or timber [31].

Mobile-based climate information services—delivered through platforms such as iCow or county weather advisories—allow farmers to access seasonal forecasts, planting advice, and market prices, enhancing their capacity to make informed decisions. Extension officers, deployed at the ward level, work closely with farmer groups, youth associations, and cooperatives to promote adaptation innovations such as solar-powered drip irrigation, composting, and weather-indexed insurance. These technologies and services reduce risk, increase productivity, and support long-term planning.

Notably, West Pokot also supports community-level planning mechanisms, such as Climate Resilience Investment Plans (CRIPs) and Community Environmental Action Plans (CEAPs), which integrate climate adaptation into local development priorities. The involvement of civil society and participatory governance structures has enhanced community ownership of adaptation strategies, contributing to more sustainable and scalable outcomes.

3.2.3.3. Transformative Capacity: Structural Change for Long-Term Resilience

Transformative capacity refers to the ability of social, ecological, and institutional systems to undertake fundamental, systemic changes that reconfigure how resources are governed, livelihoods are secured, and future risks are addressed. Unlike absorptive or adaptive responses—which focus on resisting or adjusting to shocks—transformation implies a deeper shift toward structurally different and more resilient development pathways. In agricultural contexts, this may involve rethinking land governance, policy structures, technological systems, and community engagement to sustainably confront the long-term impacts of climate change.

In Goromonzi, the potential for transformative change exists, but is currently constrained by a combination of institutional rigidity, inconsistent land reform outcomes, and weak policy enforcement. The legacy of Zimbabwe's Fast Track Land Reform Programme (FTLRP) continues to shape land relations in the district, with many smallholders occupying land without formal tenure documentation. This lack of clarity limits their ability to invest in long-term improvements or access finance and government programs, perpetuating a cycle of insecurity and low productivity [21].

Transformative capacity is further hindered by the fragmentation of agricultural governance, where overlapping responsibilities between traditional and state institutions generate confusion and weaken accountability. However, emerging initiatives signal pathways for structural change. Recent pilot programs supporting customary land registration, alongside discussions around integrating customary tenure with statutory law, suggest a movement toward a more harmonized land governance system [34]. Additionally, inclusive extension models that aim to bring services closer to marginalized groups, including women and youth, are being trialed by NGOs and local authorities.

The rise of youth-led farming cooperatives, often supported by NGOs or diaspora investments, points to a generational shift in how agriculture is perceived and practiced. These cooperatives are exploring climate-smart practices, digital marketing, and value-added processing, which not only enhance resilience but also reshape livelihood aspirations. For these changes to become transformative, however, they require state backing, sustained funding, and policy coherence across land, environment, and rural development sectors.

In contrast, West Pokot demonstrates more tangible progress toward transformative resilience, largely enabled by institutional innovations following Kenya's 2010 constitutional reforms. The introduction of devolved governance has shifted the power to plan and execute development interventions to the county level, allowing for contextualized, inclusive, and forward-looking climate strategies. One of the most transformative developments has been the implementation of the Community Land Act [35], which legally recognizes and protects communal land under customary ownership, facilitating participatory land use planning and improved natural resource management [8].

West Pokot's County government has integrated climate change into its development agenda, including the creation of Climate Resilience Investment Plans (CRIPs) and ward-level adaptation frameworks that align local priorities with broader policy goals. These plans are developed through multi-stakeholder dialogues, including elders, women's groups, youth associations, and technical experts, ensuring that local knowledge informs decision-making and implementation. This participatory governance model reflects a transformative shift in how development is planned, owned, and sustained at the grassroots level.

The promotion of climate-smart infrastructure, such as solar-powered irrigation, rangeland rehabilitation, and weather-indexed insurance, further exemplifies structural change that combines environmental sustainability with socio-economic development. Additionally, integration of indigenous knowledge systems, particularly in pastoral mobility and resource management, has helped reposition traditional institutions as legitimate actors in formal governance frameworks.

This synergy between legal reform, community engagement, and environmental planning has enabled West Pokot to move beyond incremental adaptation and toward structural transformations that improve equity, sustainability, and institutional legitimacy. However, continued challenges—including resource competition, population pressure, and regional conflict—highlight the need for continued support, monitoring, and innovation.

3.3. Data Collection and Analysis

3.3.1. Expert Consultations and Stakeholder Insights

To complement the open-source data collection, in-depth consultations were conducted with experts and stakeholders across key theoretical dimensions in Zimbabwe and Kenya. These insights were essential in validating the contextual relevance of secondary data, identifying knowledge gaps, and ensuring alignment with lived realities at community and institutional levels. The expert engagements were structured around the three guiding frameworks: Sustainable Livelihoods Framework (SLF), Institutional Theory, and Resilience Theory (see Table 1 in Appendix).

3.3.2. Expert Consultations and Stakeholder Insights Analysis: Structural Equation Modeling Approach

To determine and examine the relationships between the core theoretical con- structs (Institutional Theory, Sustainable Living Standards Framework (SLF), and Resilience Theory) based on expert perceptions. The authors use the Structural Equation Modeling (SEM) approach to do this. The conceptual model developed in this study exhibits interrelationships between the three core theoretical constructs and resilience predictors (see Figure 4). The model is used to test the following hypothesis:

- **H1**: Institutional theory positively influences the components of the SLF.
- **H2**: SLF positively influences farmer resilience to climate change.

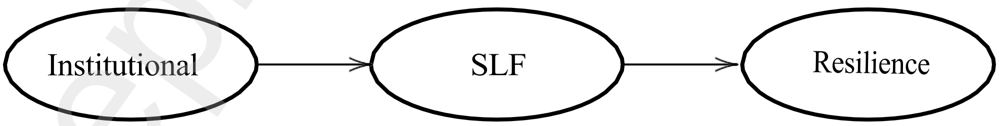

Fig. 4. Conceptual model developed to core theoretical constructs.

3.3.2.1. Survey design

The data used in this study were obtained through a questionnaire survey ad- ministered to experts from Kenya and Zimbabwe knowledgeable in agricultural adaptation and relevant policy frameworks within their respective countries. Experts targeted included those from Governmental, international organization, Academic, community-based organization. Private sector and financial institution as shown in Table 2 below:

Table 2. Experts' organization types

	Category	Kenya	Zimbabwe
1	Government	8	10
2	International Organization	2	2
3	Academic	3	5
4	Community-Based Organization	8	1
5	Health and Development Organization	6	1
6	Private Sector	0	4
7	Financial Institution	0	2
		27	25

The questionnaire consisted of 10 indicators (questions) and was divided into three sections: SLF (5 questions), Institutional Theory (2 questions), and Resilience Theory (3 questions). To ensure that the answers will be exact and clear. Each question is scored on a 1 to 10 scale, where:

- 1 = Very Poor / Not Functional
- 10 = Excellent/Fully Functional

where 1 indicates very poor performance or capacity, and 10 indicates excellent performance or full capacity Each expert should score based on their domain knowledge and experience in their specific country context.

3.3.2.2. Participants and data collection

The participants involved in the survey were experts from government, international, academic, community-based, health, development, private, and financial organizations in Kenya and Zimbabwe. The questionnaire was self- administered through an online survey system (Google Forms), which lasted for 10 days between 6th to the 16th April 2025. The experts were targeted through social media class groups, including WhatsApp and LinkedIn, which are commonly used by them in their organizations. A message comprising the objectives and importance of the survey, a link to the questionnaire and a consent to fill in the questions was sent to almost all social media platforms of student groups. Participants were encouraged to recommend the link to other coursemates and friends at the university. to the questionnaire.

3.3.2.3. Structural Equation Modelling

Structural Equation Modeling (SEM) is a set of statistical techniques commonly used for investigating the direct and indirect effects of relationships between observed and latent variables [37, 38].
In SEM, an observed variable is a variable that has been directly measured. A latent variable is a variable that cannot be directly measured but is thought to be the source of the observed variables. More precisely, it is a hypothetical, hidden variable whose values can be estimated from observable data [37-39]. In addition, the importance of SEM lies in its ability to provide a comprehensive and flexible framework for testing theories and hypotheses. On the other hand,

SEM encompasses many methods, such as exploratory factor analysis, confirmatory factor analysis, and path analysis [40, 39].
SEM is widely used in many fields such as public action on climate change [41], preventive practices [37], purchase behavior [42], agriculture [43], digital government [44], and climate and human activity [45].
The whole SEM model contains two sub-models [37, 38, 42, 46] (see Figure 5):

- The measurement model describes the relations between latent variables and their indicators (observed variables).
- The structural model describes the relations between latent variables.

The measurement model is defined as [37]:

$$x_1 = A_1\eta + \varepsilon_1$$
$$x_2 = A_2\xi + \varepsilon_2$$

Where x_1 and x_2 are random vectors of the observed variables that explain endogenous (dependent) η and exogenous (independent) ξ latent variables, A_1 and A_1 are loading matrices, and ε_1, and ε_2 are random vectors of measurement errors. The measurement model can also be expressed in matrix form as [1]:

$$Y = \begin{pmatrix} x_1 \\ X_2 \end{pmatrix} = \begin{pmatrix} A_1 & 0 \\ 0 & A_2 \end{pmatrix} \begin{pmatrix} \eta \\ \xi \end{pmatrix} + \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \end{pmatrix}$$

On the other hand, the structural model can be defined as [37]:

$$\eta = \beta\eta + \Gamma\xi + \delta$$

where β and Γ are matrices of regression coefficients that describe the cause- and-effect relationship between η and ξ, and δ is a random vector of error of endogenous variables.

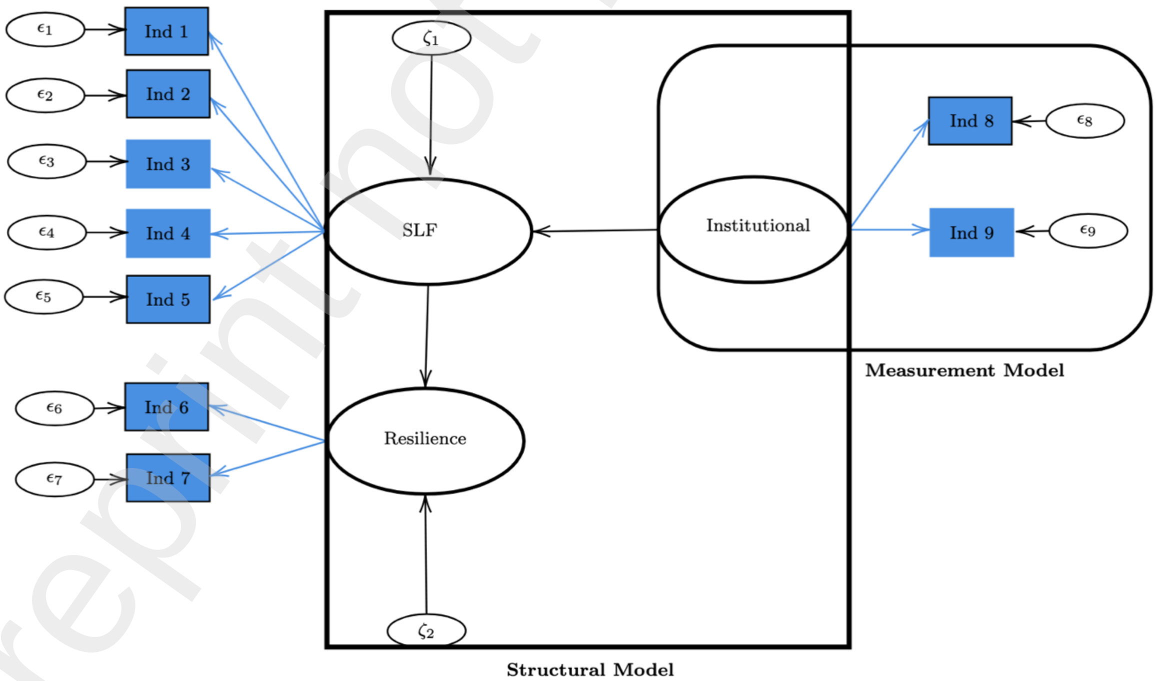

Figure 5. Schematic representation of the two models (structural and measurement).

Table 3 provides the codes and definitions for each indicator and identifies the latent variable associated with each indicator.

Table 3 Latent variables and these indicators

Latent Variable	Indicator and code
Institutional	Formal Institutions (Ind 8) Informal Institutions (Ind 9)
SLF	Natural Capital (Ind 1) Social Capital (Ind 2) Human Capital (Ind 3) Physical Capital (Ind 4) Financial Capital (Ind 5)
Resilience	Absorptive Capacity (Ind 6) Adaptive Capacity Adjustment (Ind 7)

To estimate the model, we used the PLSPM package in R software [45, 47]. Subsequently, the accuracy of the structural model fit was assessed using the coefficient of determination (R^2). The goodness-of-fit index (GOF) reflects the overall model fit. The effectiveness of the measurement model is indicated by the loadings, Cronbach's Alpha, Dillon-Goldstein's rho, and the communality index [38, 45, 47, 48].

4. Results and Discussion

Table 4 below summarizes the descriptive statistics for the indicators.

Table 4 Descriptive Statistics Summary for Indicators

Statistic	Natural Capital	Social Capital	Human Capital	Physical Capital	Financial Capital	Formal Inst.	Informal Inst.	Absorptive Capacity	Adaptive Capacity
Min.	2.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1st Qu.	3.000	4.000	3.000	2.000	2.000	3.000	2.000	2.500	4.000
Median	4.000	5.000	5.000	3.000	3.000	4.000	4.000	4.000	5.000
Mean	4.529	5.255	4.725	3.294	3.196	4.569	4.353	3.961	4.882
3rd Qu.	6.000	7.000	6.000	4.000	4.000	6.000	6.000	5.000	6.000
Max.	9.000	9.000	9.000	8.000	9.000	9.000	9.000	8.000	9.000

Table 4 shows that the range and central tendency values observed across all indicators fall within the expected scale, suggesting consistent and reliable responses from the surveyed experts. Table 5 gives the SEM model results for Kenya.

Table 5 SEM Results Summary for Kenya

Outer Model (Loadings and Communality)			
Latent Variable	**Indicator**	**Loading**	**Communality**
Institutional	Formal Institutions	0.957	0.915
	Informal Institutions	0.92	0.844
SLF	Natural Capital	0.856	0.7363
	Social Capital	0.906	0.820
	Human Capital	0.946	0.890
	Physical Capital	0.89	0.79
	Financial Capital	0.787	0.619
Resilience	Absorptive Capacity	0.872	0.76
	Adaptive Capacity Adjustment	0.864	0.746
Inner Model (Path Coefficients and Significance)			

Path	Coefficient (β)	P-value
Institutional → SLF	0.709	0.000005
SLF → Resilience	0.587	0.000126
Model Fit		
Metric	**Value**	-
R^2 (SLF)	0.802	-
R^2 (Resilience)	0.745	-
Goodness-of-Fit (GoF)	0.785	-

Table 5 shows that the evaluation of the measurement model reveals satisfactory performance. Indicator loadings, which reflect the correlation between indicators and their respective latent variable, are generally strong, with most exceeding the common threshold of 0.60 and several showing high values. In addition, the communality values are usually stronger than 0.66. This indicates that the constructs adequately explain the variance in their associated indicators.

Furthermore, the internal consistency reliability was assessed using Cronbach's Alpha (C.alpha) and Dillon-Goldstein's rho (DG.rho). All constructs demonstrated acceptable to good reliability, with DG.rho values exceeding the recommended threshold of 0.70 (Institutional: 0.762, SLF: 0.845, Resilience: 0.859), confirming the reliability of the measurement scales Regarding the structural model, SLF (R^2 = 0.802) and resilience (R^2 = 0.745) demonstrate substantial predictive accuracy. For the overall model fit, as assessed by the GoF index, it is 0.785. This indicates an excellent overall fit of the model.

Besides, the path coefficients within the model illustrate the impacts of la- tent variables, revealing that the latent variables institutional, SLF, and resilience are all significant. Table 6 below gives the model results for Zimbabwe.

Table 6 SEM Results Summary for Zimbabwe

Measurement Model (Loadings and Communality)			
Latent Variable	**Indicator**	**Loading**	**Communality**
Institutional	Formal Institutions	0.966	0.933
	Informal Institutions	0.650	0.4225
SLF	Natural Capital	0.774	0.59
	Social Capital	0.734	0.53
	Human Capital	0.863	0.74
	Physical Capital	0.878	0.772
	Financial Capital	0.869	0.75
Resilience	Absorptive Capacity	0.949	0.901
	Adaptive Capacity Adjustment	0.866	0.749
Structural Model (Path Coefficients and Significance)			
Path	**Coefficient (β)**	**P-value**	
Institutional → SLF	0.544	0.000499	
SLF → Resilience	0.659	0.00037	
Model Fit			
Metric	**Value**	-	
R^2 (SLF)	0.632	-	
R^2 (Resilience)	0.7835	-	
Goodness-of-Fit (GoF)	0.70	-	

Table 6 proves that the loadings and communality values are higher than 0.40. This indicates that the performance of the measurement model is good. Additionally, internal consistency reliability was assessed using C.alpha and DG.rho. While C.alpha for the institutional construct was slightly below the common 0.70 threshold (0.695), the more robust DG.rho met the criterion (DG.rho = 0.722); both SLF (DG.rho = 0.877) and Resilience (DG.rho = 0.908) demonstrated strong reliability. Concerning the structural model, the R^2 values of SLF and resilience are, respectively, 0.632 and 0.7835. Indicating a good predictive accuracy. Furthermore, the GOF value reached 0.75, indicating an excellent overall fit of the model. On the other hand, the path coefficients within the model illustrate the impacts of latent variables, revealing that the latent variables—institutional, SLF, and resilience—are all significant. Based on these results, Table 7 presents the Summary of hypothesis testing.

Table 7 Summary of hypothesis testing

Country	Hypothesis	Standard estimate	p-value	Decision
Kenya	H1: Institutional → SLF	0.709	<0.05	Significant
Kenya	H2: SLF → Resilience	0.587	<0.05	Significant
Zimbabwe	H1: Institutional → SLF	0.544	<0.05	Significant
Zimbabwe	H2: SLF → Resilience	0.659	<0.05	Significant

4.1. Institutional Influence on Sustainable Livelihood Framework (SLF)

Findings from Table 8 confirms that institutions significantly influence the components of the Sustainable Livelihoods Framework (SLF) in both Kenya (as shown in Figure 6) and Zimbabwe (as shown in Figure 7). In Kenya (β = 0.709), the relationship is notably stronger compared to Zimbabwe (β = 0.544). This suggests that the quality, flexibility and support of institutional structures play a critical role in shaping livelihood assets, strategies and outcomes. In Zimbabwe, the institutional landscape is largely centralized, where government control over agricultural extension services is pronounced. Extension officers are deployed and managed centrally, which may limit localized responsiveness and innovation in livelihood support [49]. Conversely, Kenya has embraced decentralization in agricultural services, enabling non-governmental organizations (NGOs) and community-based actors to work alongside government extension workers. This collaborative approach enhances access to livelihood-enabling services and knowledge dissemination [50].

4.2. Sustainable Livelihoods Framework (SLF) and Farmer Resilience

The findings indicate that Sustainable Livelihood Framework (SLF) components significantly enhance farmer resilience to climate change in both Kenya (β = 0.587) and Zimbabwe (β = 0.659). This aligns with Resilience Theory, which asserts that resilience is not solely a function of exposure to shocks but also depends on adaptive capacities rooted in social, economic, natural, human, and physical capital [51]. Notably, Zimbabwe's slightly higher resilience coefficient may be attributed to targeted post-crisis interventions, including livelihood support programs for example following the 2019 drought and Cyclone Idai [52]. Despite a highly centralized institutional framework that often delays decision-making and slows NGO project implementation due to bureaucratic protocols, Zimbabwe exhibits strong rural resilience. This is underpinned by traditional coping mechanisms, increased adoption of ICT, expanding agricultural education, market access and a rise in value addition practices among farmers. These indirect pathways are boosting resilience, even under top-down governance constraints.

In contrast, Kenya's decentralized, integrated approach with robust NGO involvement facilitates faster implementation of farmer training, financial access, and market linkages, all of which directly support resilience-building [53].

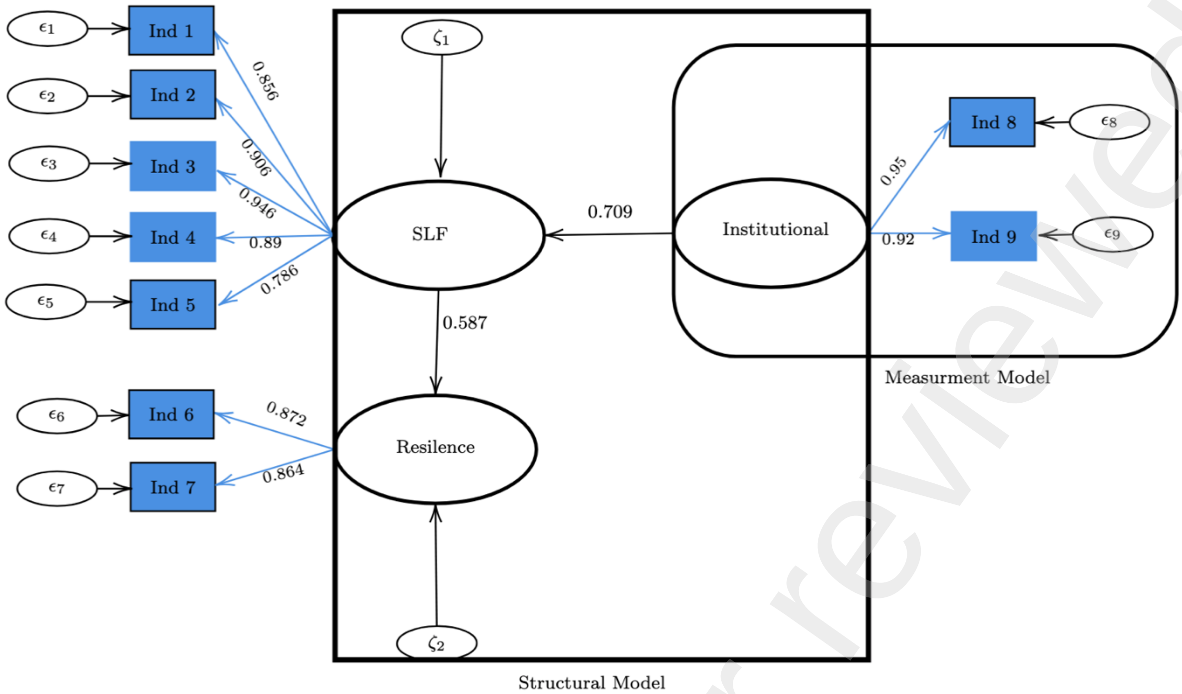

Fig 6. Model with estimated coefficients for Kenya

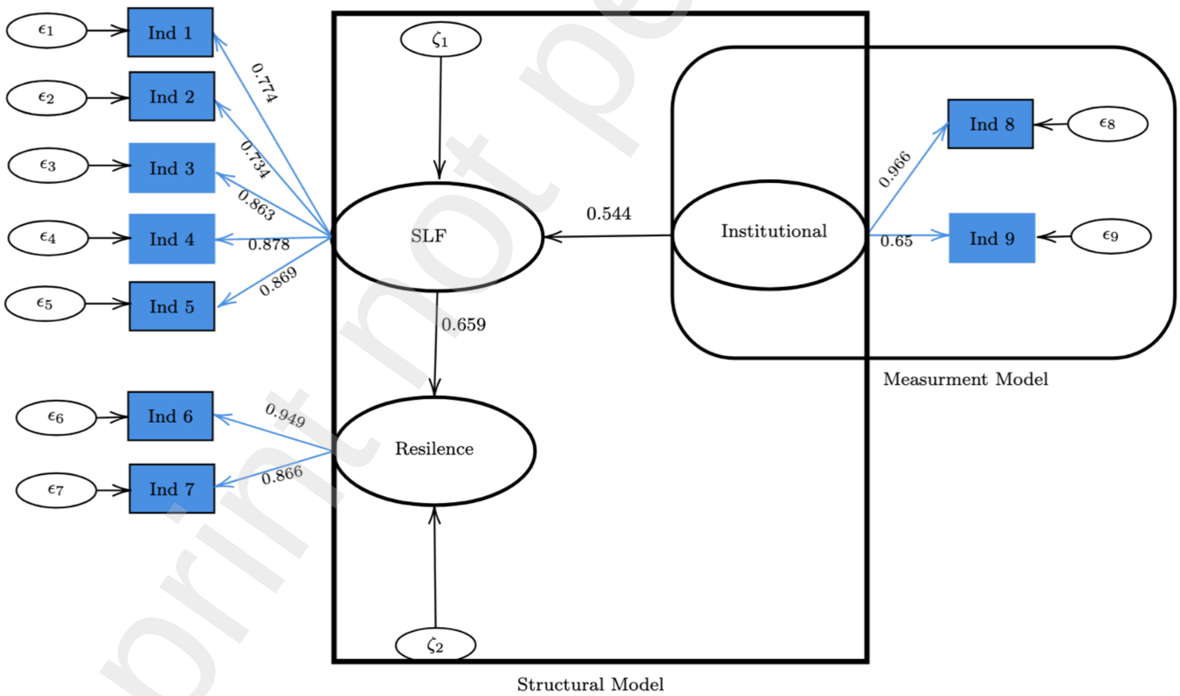

Fig 7. Model with estimated coefficients for Zimbabwe

5. Policy Implications and Recommendations

This study highlights several actionable insights for strengthening climate change adaptation in smallholder agricultural systems across sub-Saharan Africa. The comparative analysis between Kenya and Zimbabwe underscores the importance of contextualizing agricultural extension policies within decentralized or centralized governance systems, depending on institutional capacity and local conditions. These insights serve as the foundation for the subsequent recommendations, which emphasize the need for context-sensitive institutional reform, integrated extension services, inclusive financing mechanisms, and robust monitoring systems tailored to the varied realities of climate-vulnerable farming communities.

Effective governance and institutional alignment are critical to the successful implementation of climate adaptation in agriculture. In Kenya, the decentralized governance system has provided counties with significant flexibility in delivering agricultural services tailored to local conditions. County governments, often in partnership with NGOs and private sector actors, have leveraged mobile-based platforms and other innovative tools to improve agricultural outcomes. To enhance this capacity, it is essential to strengthen county-level technical expertise and financial autonomy through performance-based grants and sustained capacity development. Furthermore, institutionalizing cross-county coordination platforms will foster knowledge-sharing and ensure the harmonization of climate-smart agriculture (CSA) practices.

In contrast, Zimbabwe's centralized institutional framework offers national policy coherence but lacks sufficient flexibility to address localized challenges. There is a need to empower district-level governance structures through the establishment of innovation hubs that promote community-led experimentation. Moreover, reforms to the land tenure system are essential to provide farmers with long-term security, thereby incentivizing investment in sustainable land use and climate-resilient practices.

Extension systems remain the backbone of farmer support in both countries, playing a central role in enhancing human and knowledge capital as defined in the Sustainable Livelihoods Framework. These systems are instrumental in building the skills, competencies, and information flows that underpin adaptive agricultural decision-making under climate stress. To address the disparities between urban and rural service provision, governments should embed extension officers within local communities and incentivize field-based deployments. A nationally standardized climate-smart extension curriculum that integrates modern agricultural technologies with indigenous knowledge will provide a strong foundation for adaptation. Public-private partnerships are crucial for scaling digital extension platforms and expanding access to real-time climate forecasting services.

On the issue of land tenure and resource access, both countries must reconcile the coexistence of formal statutory laws with informal customary arrangements. In Kenya, the Community Land Act [35] should be actively enforced to secure land rights for pastoralist and communal land users. Zimbabwe must similarly integrate customary tenure into its formal legal system to reduce conflicts and unlock long-term investment in agriculture.

Climate finance and investment in resilience must be expanded at the grassroots level. Establishing community-managed micro-adaptation funds with clear and transparent access criteria will empower local farmer groups. Bundled solutions that combine insurance, credit, and advisory services will offer a more holistic approach to managing climate risks, particularly for vulnerable crops such as sweet potatoes. Engagement with local financial institutions, including SACCOs and fintech companies, will be critical in promoting financial inclusion.

To track adaptation outcomes effectively, robust monitoring, learning, and evaluation (MEL) systems must integrate resilience metrics directly into agricultural performance frameworks. This includes harmonizing data collection across ministries and national statistical agencies to ensure coherence and comparability. Additionally, investments should be made in localized climate risk assessments and vulnerability mapping to provide targeted insights that guide interventions at both national and sub-national levels. Governments should incorporate resilience metrics into existing agricultural performance frameworks and harmonize data collection across ministries and national statistics offices. Investments in localized climate risk assessments and vulnerability mapping will help target interventions more effectively.

To achieve policy coherence, sectoral strategies such as Kenya's National Adaptation Plan (NAP), the Agricultural Sector Development Strategy (ASDS), and the National Climate Change Action Plan (NCCAP) should be aligned with devolved planning frameworks. Similarly, Zimbabwe's implementation of the ZUNDAF [54] must be anchored in clear operational guidelines that address agricultural adaptation.

Nutritional outcomes should also be integrated into agricultural planning. Building on Zimbabwe's National Nutrition Strategy, extension services should promote nutrition-sensitive agriculture, particularly through the support of sweet potato value chains. Developing local processing enterprises and aggregation hubs will enhance value addition and improve food security.

Finally, capacity development and social inclusion are essential for long-term resilience. Training female extension officers and empowering them as lead facilitators in climate adaptation will ensure gender-responsive programming. Youth engagement must also be prioritized through the promotion of ICT-based agribusinesses and climate entrepreneurship. Ensuring that all adaptation efforts are inclusive and equitable will be central to their sustainability and success.

This study highlights several actionable insights for strengthening climate change adaptation in smallholder agricultural systems across sub-Saharan Africa. The comparative analysis between Kenya and Zimbabwe underscores the importance of contextualizing agricultural extension policies within decentralized or centralized governance systems, depending on institutional capacity and local conditions. These insights serve as the foundation for the subsequent recommendations, which emphasize the need for context-sensitive institutional reform, integrated extension services, inclusive financing mechanisms, and robust monitoring systems tailored to the varied realities of climate-vulnerable farming communities.

6. Conclusion and Perspectives

This study highlights how agricultural extension, land governance, and community adaptation shape resilience in smallholder sweet potato farming in Kenya and Zimbabwe. Using SEM and guided by the Sustainable Livelihoods Framework (SLF), Institutional Theory, and Resilience Theory, the research demonstrates the central role of institutional quality, resource access, and grassroots innovation in enabling adaptation. Kenya's decentralized model fosters flexibility and digital innovation, while Zimbabwe's centralized approach provides policy coherence but needs more local responsiveness. The findings show that resilience emerges from the alignment of formal policy, community agency, and continuous learning, not from infrastructure alone. The study underscores the need for hybrid governance systems that combine top-down coordination with bottom-up participation. It recommends aligning climate finance, land tenure reforms, and extension policies to local realities, with an emphasis on inclusive and gender-responsive strategies.

While this study provides valuable comparative insights for Kenya and Zimbabwe, certain limitations shape perspectives for future research. Firstly, the geographical scope is confined to these two countries; thus, caution is needed when generalizing findings across diverse sub-Saharan African contexts. Secondly, the 'Institutional' and 'Resilience' constructs were measured using only two indicators each, potentially limiting measurement richness. Ongoing work aims to address these aspects: (1) extending the geographical scope to encompass a wider range of national contexts, and (2) employing more comprehensive, validated multi-item scales for the core theoretical constructs in subsequent analyses.

Finally, Future research should focus on three core themes: scaling insights across crops and regions, linking climate finance with social protection and aggrotech, and deepening the integration of youth and gender into adaptation planning.

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Appendix

Table 1. The study area dynamics, indexes, data needed, sources and Targeted experts

Study Area Dynamics		**Indexes**	**Country**	**Data needed**	**Targeted Experts**
Sustainable Livelihoods Framework (SLF)	1	Natural Capital	Zimbabwe	Geographic data on climate-vulnerable zones, migration corridors, land degradation, drought frequency	Environmental scientists, GIS analysts, local land officers
			Kenya	Arid and semi-arid zones (ASALs), water stress maps, deforestation trends, soil fertility data	Ministry of Agriculture officials, ecologists, climate change focal points
	2	Social Capital	Zimbabwe	Role of social networks in migration, diaspora linkages, community integration of migrants, traditional leadership influence.	Local leaders, traditional chiefs, community-based organization (CBO) reps
			Kenya	Inter-ethnic cohesion, host-refugee relations (e.g., in Turkana/Dadaab), role of churches and women's groups in supporting migrants.	NGO practitioners, ward admins, social development officers
	3	Human Capital	Zimbabwe	Education levels of migrants, skill transferability, access to health and education, migration training and reintegration programs.	Labour economists, teachers, vocational trainers, health workers

			Kenya	Labour migration trends, youth employment stats, refugee education enrollment, vocational training reach, climate literacy	Technical and Vocational Education and Training (TVET) officers, UNHCR livelihoods focal persons
	4	Physical Capital	Zimbabwe	Transport networks in migrant areas, housing and health infrastructure, communications technology, energy access.	Urban planners, infrastructure ministry officials, local engineers
			Kenya	Access to markets, ICT, rural electrification, health facility distribution in migration-prone zones	Infrastructure development agencies, planners, energy officers
	5	Financial Capital	Zimbabwe	Migration-related remittances, financial inclusion of migrants, access to credit, informal savings and lending systems.	Central bank officials, mobile money providers, microfinance institutions
			Kenya	Use of mobile money (e.g., M-Pesa) among migrants/refugees, diaspora bonds, national migration finance policies.	Fintech experts, banking regulators, humanitarian cash transfer program leads
Institutional Theory	6	Formal Institutions and Access to Resources	Zimbabwe	National and local land policies, extension service frameworks, climate adaptation policies, government-led migration programs.	Policy analysts, extension officers, government planners
			Kenya	Implementation of Community Land Act (2016), decentralized governance reports, county climate adaptation plans.	County government planners, land officers, legal scholars
	7	Informal Institutions and Land Tenure Governance	Zimbabwe	Customary land access rules, role of traditional leaders, local dispute resolution records, inheritance practices.	Chiefs, headmen, legal anthropologists
			Kenya	Clan-based land access systems, councils of elders, customary pastoral mobility rules.	Elders, local chiefs, community mediators

Resilience Theory	8	Absorptive Capacity: Withstanding Climate Shocks	Zimbabwe	Disaster preparedness plans, access to emergency aid, drought relief history, communal risk-sharing mechanisms.	Disaster risk officers, NGO emergency teams, local CSOs
			Kenya	Early warning systems, drought contingency plans, water trucking records, local disaster coping strategies	County disaster managers, humanitarian coordinators
	9	Adaptive Capacity: Adjusting to Climatic Variability	Zimbabwe	Conservation agriculture uptake, adoption of drought-resistant crops, farmer training programs, irrigation use.	Agronomists, climate-smart agriculture trainers, extension agents.
			Kenya	Uptake of agroforestry, mobile-based agro-advisories, integration of traditional and modern farming practices.	Agripreneurs, NGO trainers, agro-extension specialists
	10	Transformative Capacity: Structural Change for Long-Term Resilience	Zimbabwe	Land tenure reform progress, inclusive agricultural policy shifts, youth-led agricultural innovation, decentralization trends.	Policy thinks tanks, youth farming cooperatives, agricultural innovation hubs
			Kenya	Implementation of devolved climate governance, participatory climate planning (e.g., CRIDPs), recognition of communal land rights.	County climate officers, participatory planning consultants, community-based organizations